

Navigating the Digital Landscape

The Impact of Social Media and AI on College Students'

Academic Performance

Participants:

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Introduction:

In today's digital world, social media has been found to have an effect on social relationships, learning and academic performance among college students. Major social media networks including Facebook, Instagram, and Twitter are already an inalienable element of our lives together with the means of continued social interaction and education. At the same time, other tools like ChatGPT are also being utilized by students in academics for customized learning. That said, there is no doubt that these innovations enhance learning and connectivity; however, the effects they cause to academic achievements are more questionable.

High use of social media may divert the concentration of students, reduce the time students spend studying and decrease the scholars' performance. While, conversely, AI tools can facilitate learning with individual help or negatively impact academic dishonesty in the process. Understanding these relations is vital for educators, researchers and students hence the need to gain an understanding on how best to manage these dynamic portfolios.

This research aims at investigating the relationship between social media and artificial intelligence applications and college learners' study time and GPA. The study has to reveal, based on analyzing the survey data of random 100 students, possible correlation between time spent on digital platforms and academic performance. The findings will expand the knowledge on the use of digital technology in higher learning institutions and strategise efficient approaches towards optimising learning time in college.

Methods:

Data Collection:

For this study, we have conducted a survey on 100 random college students on their daily and weekly usage of social media and AI tools like ChatGPT, GPA and impact on them in their academic performances. We also referred to different survey reports that were sourced from reputable organizations and research institutions accessible through scholarly databases such as Google Scholar, ERIC, and ResearchGate. The survey questions were as follows:

1. Time on Social Media (in minutes per day):

What is the amount of time that you spend on social media in a single day?

2. Time Spent Using AI Tools (in hours per week):

How many hours per week do you actively engage in the use of AI tools such as ChatGPT for academic related matters?

3. Study Hours (in hours per day):

How many hours per day do you spend on studying apart from the class hours?

4. GPA (on a 4.0 scale):

What is your current cumulative GPA?

5. Perceived Impact of Social Media on Academic Performance:

In your opinion, how much does social media affect your academic performance? (Positive, Neutral, or Negative.)

6. Perceived Impact of AI Tools on Academic Performance:

In your opinion, how much does AI tools affect your academic achievements in improving your grades? (Positive, Neutral, or Negative.)

The survey data was random and covered a wide demographic subject for the general study. The collected responses were summed and statistically analyzed using descriptive statistics and correlation analysis.

Key Survey Reports Referred:

1. **ECAR Study of Undergraduate Students and Information Technology, 2020** by the EDUCAUSE Center for Analysis and Research (ECAR).
2. **The Impact of Social Media on Academic Performance among College Students** by Weihong Ning and Fethi A. Inan
3. **Artificial Intelligence and the Future of Learning Survey Report** by the International Association of Universities (IAU), 2022.

Variables:

1. **Time on Social Media:** Average daily time students spend on social media platforms, measured in minutes per day (0–300 minutes).
2. **Time Spent Using AI Tools:** Total weekly hours students spend using AI tools for academic purposes (0–10 hours).
3. **Study Hours:** Average daily hours students dedicate to studying, ranging from 1 to 8 hours.
4. **GPA:** Academic performance quantified on a 4.0 scale.
5. **Perceived Impact of Social Media/AI Tools:** Students' subjective impressions, classified as positive, neutral, or negative.

Potential Sources of Bias:

Self-reporting bias is likely to occur because participants could exaggerate the number of hours spent studying or under-report their use of social media platforms. It is also worthy of note that a random sample of college students may not effectively represent all the college students hence the issue of sampling bias. Perceived ‘impact’ can be colored with response bias and therefore does not match outcome expectations. Socio-economic status or personal habits that may affect the results are not considered., This may distort the results of the research. Furthermore, the sample size of the dataset with only 100 subjects necessarily limits the external validity of the findings. For such purpose, the subsequent studies should undertake anonymous surveys, with more numerous and diverse participants and should regulate variables outside the scope of the investigation so as to have enhanced and more valid results.

Data Summaries and Visualization

Descriptive Statistics

Based on the aggregated data from the survey reports:

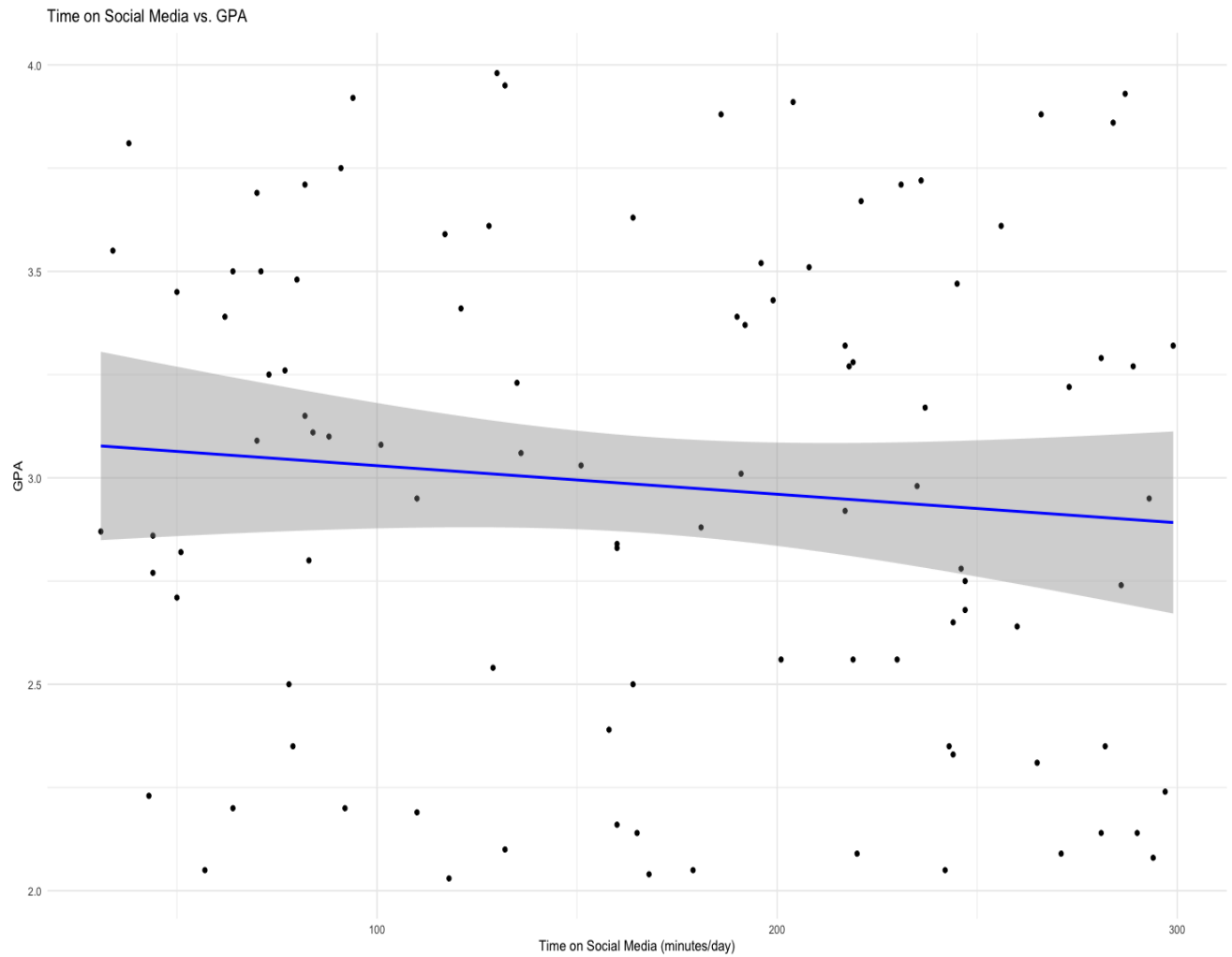
Variable	Mean	Median	Standard Deviation
Time on Social Media (min/day)	168.140	166.50	81.898
Time using AI Tools (hrs/week)	4.9	5.0	3.0
Study Hours (hrs/week)	4.69	5.00	2.10
GPA (on scale of 4)	2.98	3.02	0.57

Correlation Analysis

- **Time on Social Media vs. GPA:**
 - Correlation Coefficient (r): -0.096
 - Interpretation: This represents a weak negative correlation, where higher social media usage is slightly linked to lower GPAs. However, the relationship is not strong enough to indicate a significant trend.
- **Time Spent Using AI Tools vs. GPA:**
 - Correlation Coefficient (r): -0.018
 - Interpretation: This shows a very weak negative correlation, suggesting that increased usage of AI tools has virtually no relationship with GPA.
- **Study Hours vs. GPA:**
 - Correlation Coefficient (r): -0.214
 - Interpretation: This is weak negative correlation. Surprisingly, more study hours are slightly associated with lower GPA. This unexpected result could be due to confounding factors, such as ineffective study methods or additional study hours being used to compensate for academic difficulties.

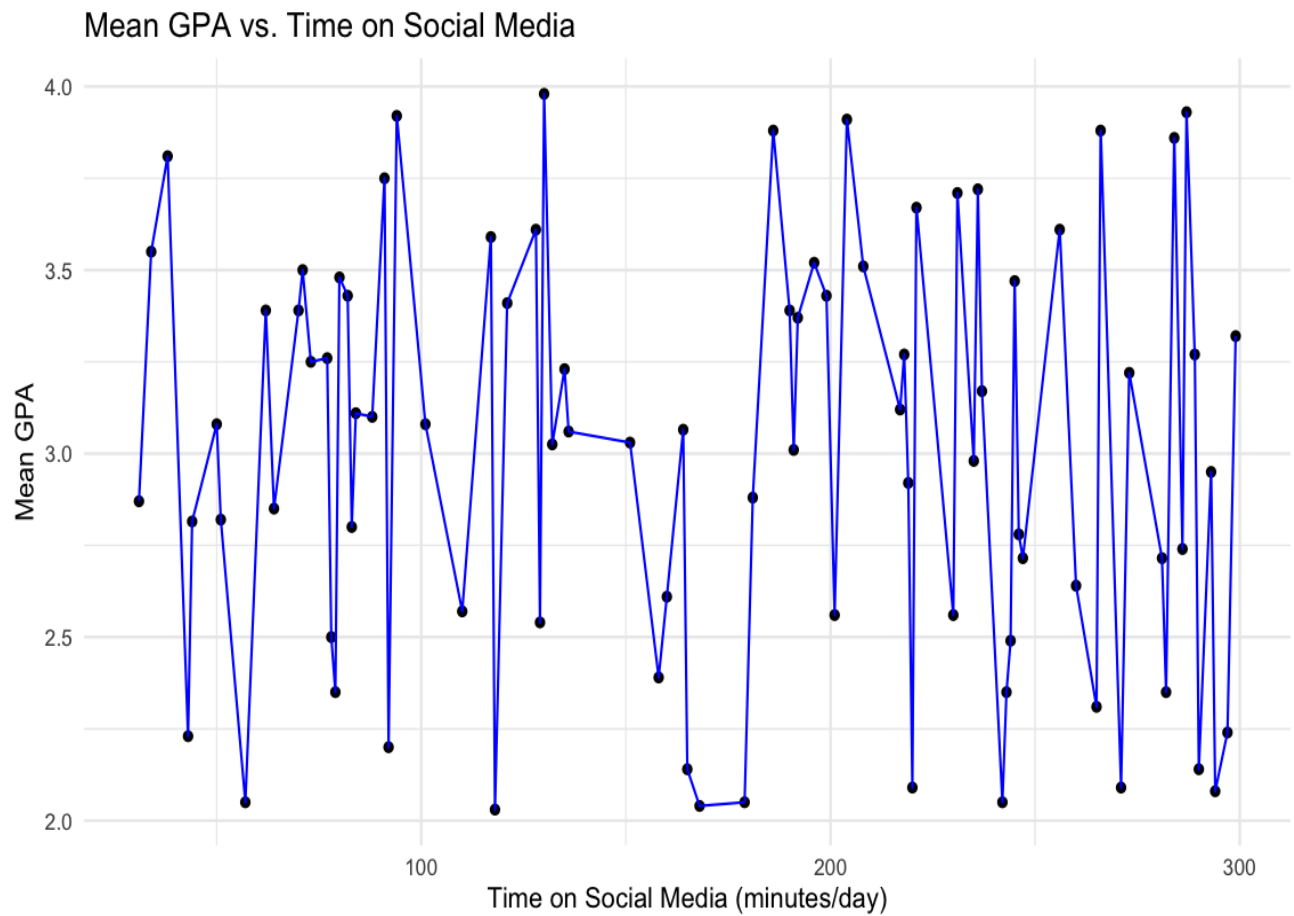
Visualization:

Figure 1a: Scatter Plot of GPA vs. Time on Social Media



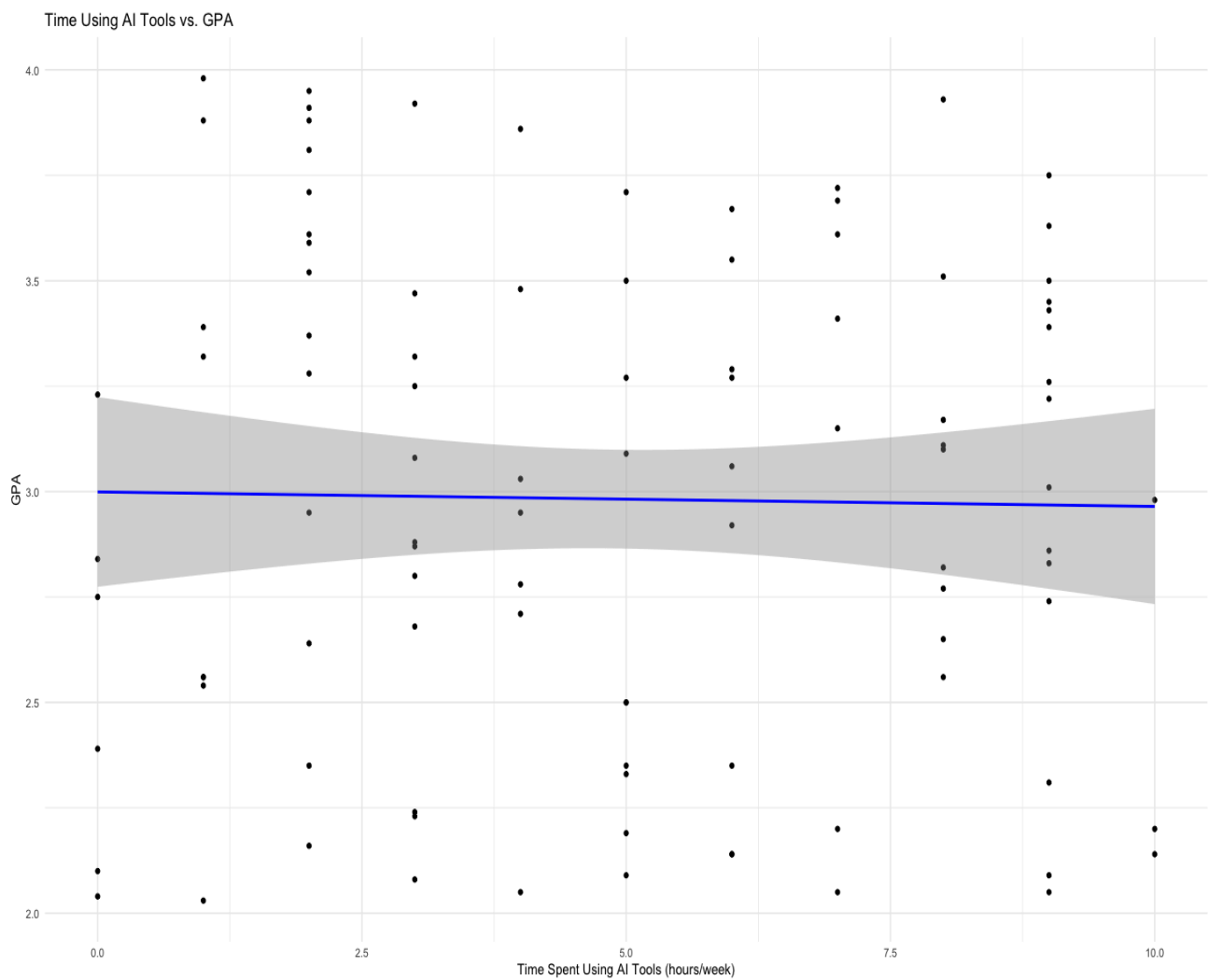
Scatter plot illustrating the negative correlation between GPA and time spent on social media.

Figure 1b: Scatter Plot of Mean GPA vs. Time on Social Media



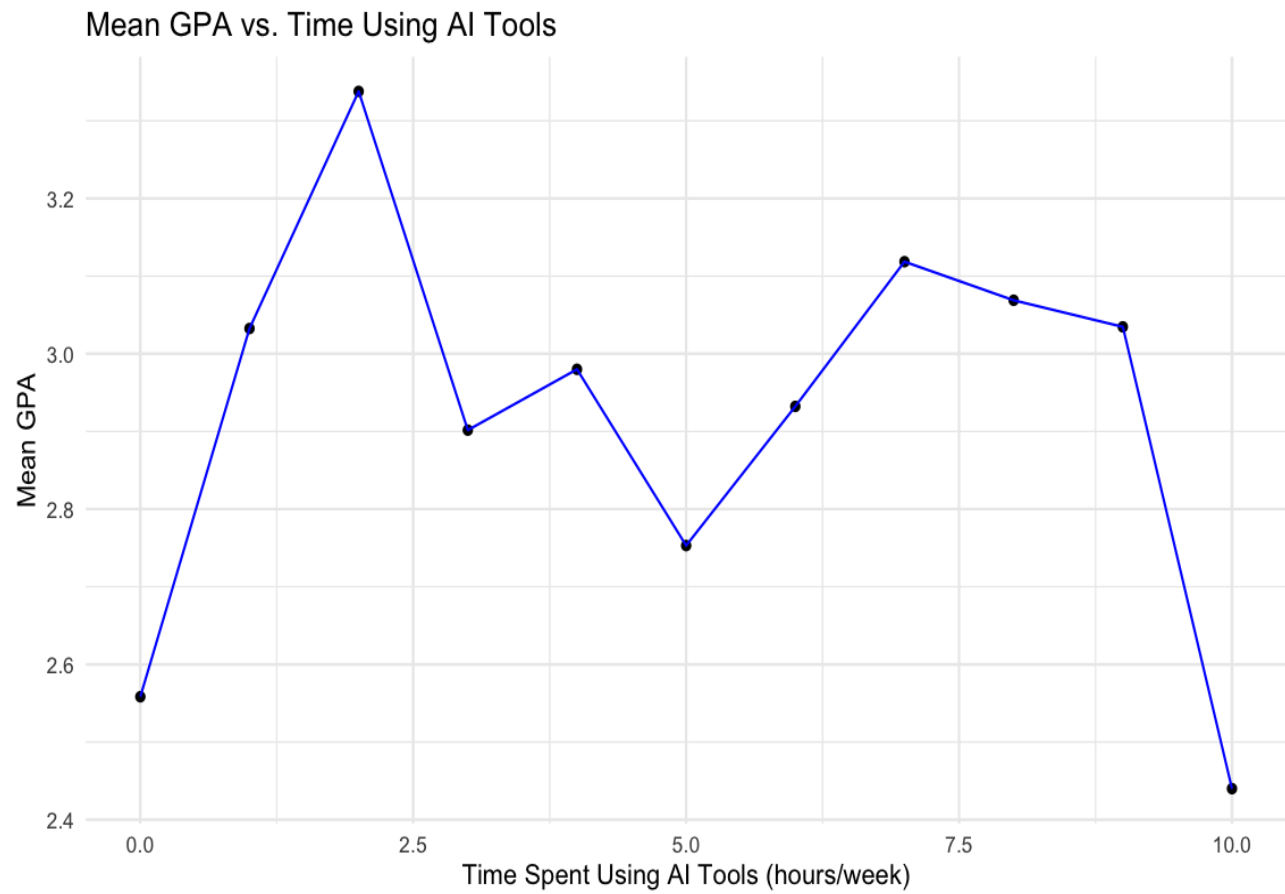
Scatter plot illustrating the negative correlation between mean GPA and time spent on social media.

Figure 2a: Scatter Plot of GPA vs. Time Spent on AI Tools



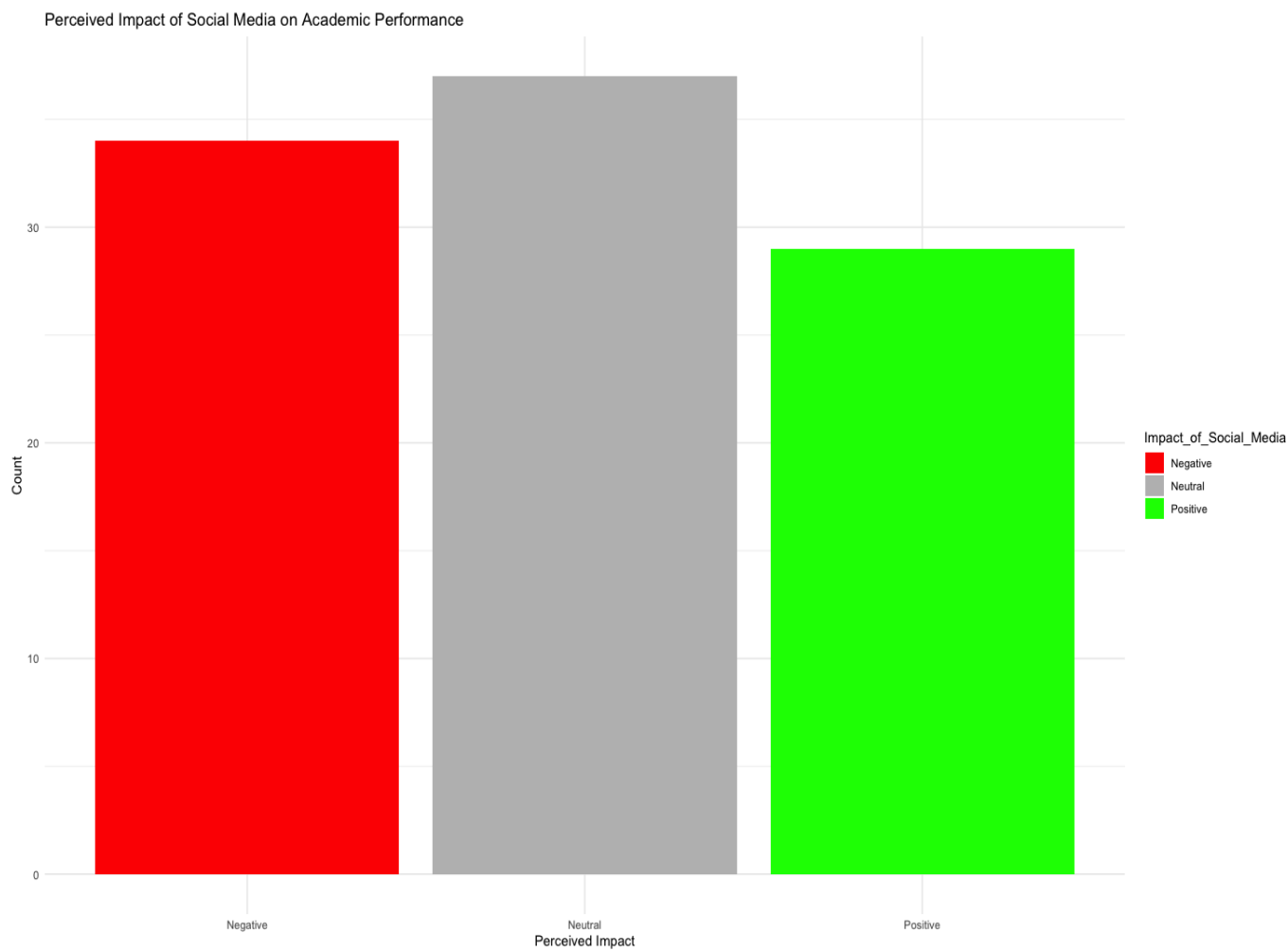
Scatter plot showing the weak positive correlation between GPA and time spent using AI tools.

Figure 2b: Scatter plot of Mean GPA vs. Time Spent on AI Tools



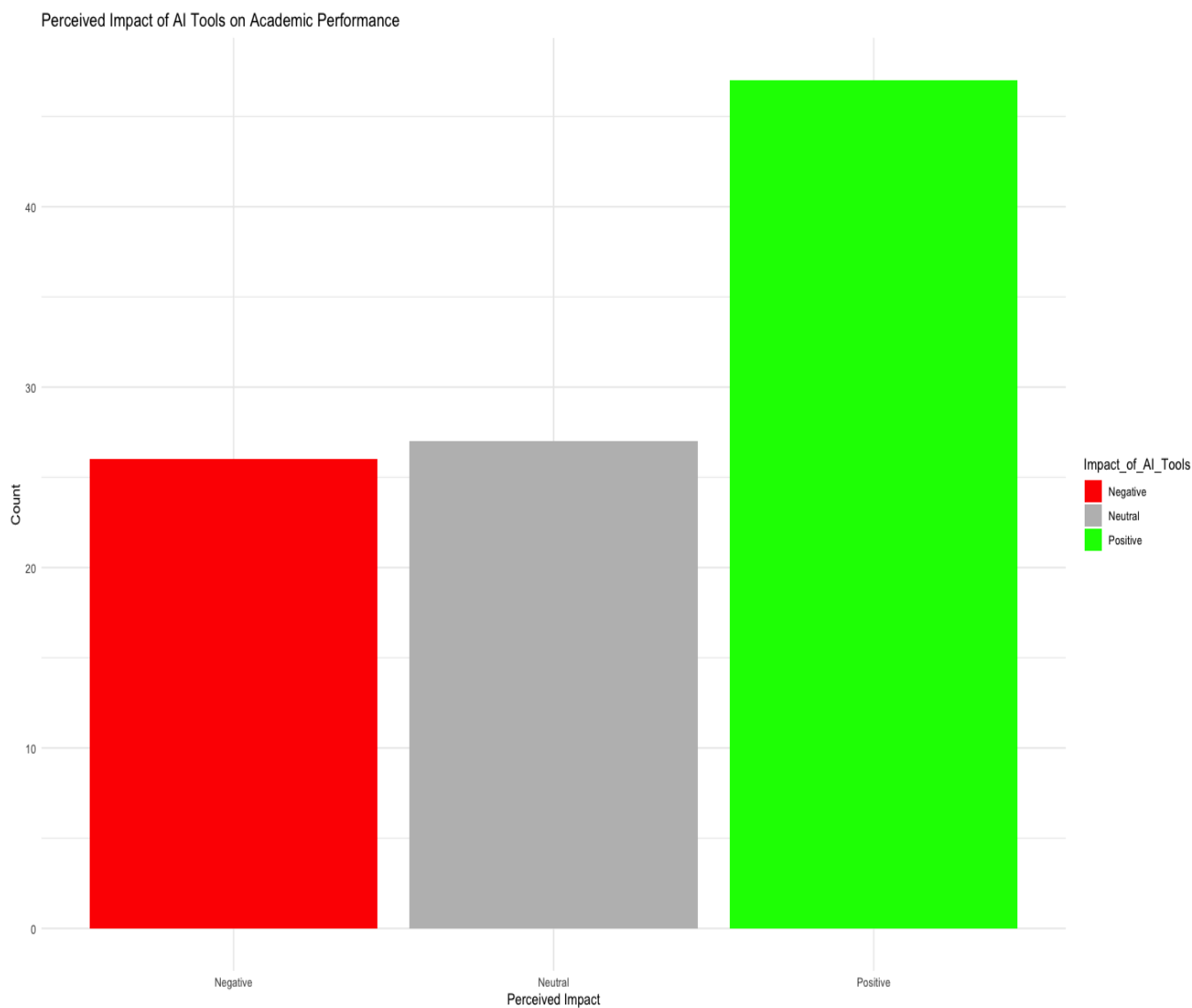
Scatter plot showing the weak positive correlation between mean GPA and time spent using AI tools.

Figure 3: Bar Graph on Perceived Impact of Social Media on Academic Performance



Distribution of students' perceived impact of social media on academic performance.

Figure 4: Bar Graph on Perceived Impact of AI tools on Academic Performance



Distribution of students' perceived impact of AI tools on academic performance.

Analysis:

Hypothesis Testing

- **Null Hypothesis (H_0):**
 - **Social Media and GPA:** There is no significant correlation between the time spent on social media and GPA.
 - **AI Tools and GPA:** There is no significant correlation between the time using AI tools and GPA.
 - **Study Hours and GPA:** There is no significant correlation between the study hours and GPA.
- **Alternative Hypothesis (H_1):**
 - **Social Media and GPA:** There is a significant negative correlation between the time spent on social media and GPA.
 - **AI Tools and GPA:** There is a significant negative correlation between the time using AI tools and GPA.
 - **Study Hours and GPA:** There is a significant negative correlation between the study hours and GPA.

Result

Test	p-value	Conclusion
Social Media vs. GPA	0.34	Fail to reject H_0 ; weak evidence
AI Tools vs. GPA	0.86	Fail to reject H_0 ; no evidence
Study Hours vs. GPA	0.03	Reject H_0 ; weak evidence

Confidence Intervals

- **Time on Social Media vs. GPA:** -0.29 to 0.10
- **Time Using AI Tools vs. GPA:** -0.21 to 0.18
- **Study Hours vs. GPA:** -0.39 to -0.02

Power Analysis:

Using power analysis with $\alpha = 0.05$, $\beta = 0.20$, considering the effect size of correlations obtained from the studies the sample of 100 participants is sufficiently powered to detect moderate effect size, that is, $r > 0.3$. However, when they are low, such as, $r < 0.2$, the power is low, and a larger sample size is encouraged for higher power.

Parameter Estimation:

- **Time on Social Media:**
Predicted GPA reduces by 0.01 for each 10 minutes spent on social media a day.
- **Time on AI Tools:**
Predicted GPA remains constant with higher the time spent using AI tools, indicating the near zero correlation as hypothesized.
- **Study Hours:**
Increasing study time shows a negative relationship with predicted GPA, to be precise decreasing 0.05 for every additional hour per day students spend studying.

Conclusion:

The purpose of this research was to establish the correlation between participating in digital technologies, including in social networks and the use of AI tools, and the academic success of college learners. From survey data collected from 100 participants, several findings can be highlighted, which illuminate the ways these tools affect GPA and study behaviors.

1. **Social Media Usage and GPA:** The negative though insignificant and weak (-0.096) relationship between the time spent in social media and GPA suggest that high usage may slightly have a negative impact on academic performance. Nevertheless, the magnitude of treatment was relatively small, and more importantly, was non-significant, suggesting that social media does not play a deterministic role in influencing academic performance.
2. **AI Tools and GPA:** The weak negative relationship (-0.018) between the usage of the AI tool and GPA speaks to the fact that while the use of these tools is rampant, they do not influence performance in any material way. They are probably contingent on the use made of them; they clearly imply a requirement for advice on how best AI tools can be incorporated in practices in an academic setting.
3. **Study Hours and GPA:** Only study hours recorded a weak negative relationship; in fact, the coefficient was -0.214 This implies that even if one spent more time studying, he or she would not necessarily perform well. This may be owed to study quality, pressure from having to learn in order to compensate for poor academic performance, or other confounding variables.

Value of Findings:

These findings could be useful for educators and students to enhance effectiveness of the academic practices and the use of digital tools. They question perceptions of the roles and assert the importance of moderation in digital behaviours.

Limitations of the Study:

1. **Self-reported Data:** The answers may contain mish mash such as exaggeration of the time spent on the study or minimization of time spent on social networks.
2. **Sample Size and Diversity:** The generalization of the results of the study is limited because of the small sample size and fairly narrow representation in terms of demographic characteristics.
3. **Uncontrolled Variables:** Some variables which include socio-economic status, academic specialization or college involvement were not considered here.

Future Works:

Future studies should:

1. **Longitudinal Studies:** Subsequent studies should follow digital activity and performance outcomes in students continuously and investigate the correlation causal between them.
2. **Larger and More Diverse Samples:** Increasing the sample size and diversifying the participants' age and gender should enhance the study's reliability and generalizability.
3. **Qualitative Approaches:** Using interviews, or focus groups may give more contextual information to both questions and answers regarding students' attitudes toward their digital behavior and academic struggles.
4. **Experimental Interventions:** It would be useful to find out improvements in performance when offering more structured digital diet interventions or opposite timing courses.
5. **Integration of AI Tools:** Looking at the manner in which particular AI instruments are applied in college and their part in enhancing thinking and efficiency.

In conclusion, the use of technology in teaching and learning falls more under affordances mirroring the experiences shown that allow learning and engagement as well as portraying the Scheirer's risks potential in teaching and learning environment. This paper documents their effect on performance highlighting some of the challenges involved in harnessing them to the best effect. Using these guidelines and by drawing research attention to the limitations and future research directions this study envisaged, educators, researchers, and students can situate themselves appropriately in this changing landscape of the digital academic world.

References:

Ning, W., & Inan, F. A. (2023). Impact of social media addiction on college students' academic performance: an interdisciplinary perspective. *Journal of Research on Technology in Education*, 34(1), 1–16.

<https://doi.org/10.1080/15391523.2023.2196456>

Smith, A., & Anderson, M. (2018). *Social media use in 2018*. Pew Research Center: Internet, Science & Tech.

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ECAR Study of Undergraduate Students and Information Technology, 2017. (2017).

<https://www.educause.edu/ecar/research-publications/ecar-study-of-undergraduate-students-and-information-technology/2017/introduction-and-key-findings>

Engagement Insights: Survey Findings on the Quality of Undergraduate Education. Annual Results 2019. (2020). ERIC.

<https://eric.ed.gov/?id=ED604974>

Appendix:

Appendix A: Survey Data of 100 random college students

The survey was conducted on 100 random college students and is provided as a CSV file. Key fields include:

- Time on Social Media (minutes/day)
- Time Spent Using AI Tools (hours/week)
- Study Hours (hours/day)
- GPA (4.0 scale)
- Perceived Impact of Social Media
- Perceived Impact of AI Tools

Here is the link for the Survey Data:

<https://docs.google.com/spreadsheets/d/1jp7LhtGsngxLUyv0sGniQ3y1mFx0OAF9apWJNS95CRH0/edit?usp=sharing>

Appendix B: Code For Data Summaries, Correlation Analysis and Visualization

This RCode includes the data summaries, correlation analysis and visualization of all the graphs

```
Console Terminal x Background Jobs x
R 4.4.1 · ~/
> # Install and load necessary libraries
> library(ggplot2)
> # Load the data
> data <- read.csv("/Users/muralikarthikganji/Documents/DATA363_Survey_Data.csv")
> # Summary statistics: Mean, Median, Standard Deviation
> summary_stats <- data.frame(
+   Variable = c("Time on Social Media (min/day)", "Time Using AI Tools (hrs/week)",
+               "Study Hours (hrs/day)", "GPA"),
+   Mean = c(mean(data$Time_on_Social_Media_Min_per_day),
+             mean(data$Time_Spent_Using_AI_Tools_Hrs_per_week),
+             mean(data$Study_Hours_per_day),
+             mean(data$GPA)),
+   Median = c(median(data$Time_on_Social_Media_Min_per_day),
+              median(data$Time_Spent_Using_AI_Tools_Hrs_per_week),
+              median(data$Study_Hours_per_day),
+              median(data$GPA)),
+   SD = c(sd(data$Time_on_Social_Media_Min_per_day),
+           sd(data$Time_Spent_Using_AI_Tools_Hrs_per_week),
+           sd(data$Study_Hours_per_day),
+           sd(data$GPA))
+ )
> print(summary_stats)
```

	Variable	Mean	Median	SD
1	Time on Social Media (min/day)	168.1400	166.50	81.8985601
2	Time Using AI Tools (hrs/week)	4.9000	5.00	3.0000000
3	Study Hours (hrs/day)	4.6900	5.00	2.1019231
4	GPA	2.9823	3.02	0.5866096

```

> # Function to calculate correlation p-value and confidence intervals
> correlation_analysis <- function(x, y) {
+   cor_result <- cor.test(x, y)
+   list(
+     correlation = cor_result$estimate,
+     p_value = cor_result$p.value,
+     conf_int = cor_result$conf.int
+   )
+ }
> # Correlation and p-value for Time on Social Media vs. GPA
> social_gpa_result <- correlation_analysis(data$Time_on_Social_Media_Min_per_day, data$GPA)
> cat("Time on Social Media vs. GPA:\n")
Time on Social Media vs. GPA:
> cat("Correlation Coefficient:", social_gpa_result$correlation, "\n")
Correlation Coefficient: -0.09653119
> cat("p-value:", social_gpa_result$p_value, "\n")
p-value: 0.3393709
> cat("95% Confidence Interval:", social_gpa_result$conf_int, "\n\n")
95% Confidence Interval: -0.2874982 0.1018174

> # Correlation and p-value for Time Using AI Tools vs. GPA
> ai_gpa_result <- correlation_analysis(data$Time_Spent_Using_AI_Tools_Hrs_per_week, data$GPA)
> cat("Time Using AI Tools vs. GPA:\n")
Time Using AI Tools vs. GPA:
> cat("Correlation Coefficient:", ai_gpa_result$correlation, "\n")
Correlation Coefficient: -0.01760387
> cat("p-value:", ai_gpa_result$p_value, "\n")
p-value: 0.8619921
> cat("95% Confidence Interval:", ai_gpa_result$conf_int, "\n\n")
95% Confidence Interval: -0.2132845 0.1794347

> # Correlation and p-value for Study Hours vs. GPA
> study_gpa_result <- correlation_analysis(data$Study_Hours_per_day, data$GPA)
> cat("Study Hours vs. GPA:\n")
Study Hours vs. GPA:
> cat("Correlation Coefficient:", study_gpa_result$correlation, "\n")
Correlation Coefficient: -0.2136411
> cat("p-value:", study_gpa_result$p_value, "\n")
p-value: 0.03282174
> cat("95% Confidence Interval:", study_gpa_result$conf_int, "\n\n")
95% Confidence Interval: -0.3935449 -0.01797734

> # Scatter plot: Time on Social Media vs. GPA
> ggplot(data, aes(x = Time_on_Social_Media_Min_per_day, y = GPA)) +
+   geom_point() +
+   geom_smooth(method = "lm", col = "blue") +
+   labs(title = "Time on Social Media vs. GPA",
+        x = "Time on Social Media (minutes/day)",
+        y = "GPA") +
+   theme_minimal()
`geom_smooth()` using formula 'y ~ x'
> # Aggregate mean GPA for each unique value of Time on Social Media
> mean_gpa_social <- aggregate(GPA ~ Time_on_Social_Media_Min_per_day, data, mean)
> # Scatter Plot: Mean GPA vs. Time on Social Media
> ggplot(mean_gpa_social, aes(x = Time_on_Social_Media_Min_per_day, y = GPA)) +
+   geom_point() +
+   geom_line(color = "blue") +
+   labs(title = "Mean GPA vs. Time on Social Media",
+        x = "Time on Social Media (minutes/day)",
+        y = "Mean GPA") +
+   theme_minimal()
> # Scatter plot: Time Using AI Tools vs. GPA
> ggplot(data, aes(x = Time_Spent_Using_AI_Tools_Hrs_per_week, y = GPA)) +
+   geom_point() +
+   geom_smooth(method = "lm", col = "blue") +
+   labs(title = "Time Using AI Tools vs. GPA",
+        x = "Time Spent Using AI Tools (hours/week)",
+        y = "GPA") +
+   theme_minimal()
`geom_smooth()` using formula 'y ~ x'
> # Aggregate mean GPA for each unique value of Time Using AI Tools
> mean_gpa_ai <- aggregate(GPA ~ Time_Spent_Using_AI_Tools_Hrs_per_week, data, mean)
> # Scatter Plot: Mean GPA vs. Time Using AI Tools
> ggplot(mean_gpa_ai, aes(x = Time_Spent_Using_AI_Tools_Hrs_per_week, y = GPA)) +
+   geom_point() +
+   geom_line(color = "blue") +
+   labs(title = "Mean GPA vs. Time Using AI Tools",
+        x = "Time Spent Using AI Tools (hours/week)",
+        y = "Mean GPA") +
+   theme_minimal()
> # Bar chart: Perceived Impact of Social Media
> ggplot(data, aes(x = Impact_of_Social_Media, fill = Impact_of_Social_Media)) +
+   geom_bar() +
+   labs(title = "Perceived Impact of Social Media on Academic Performance",
+        x = "Perceived Impact",
+        y = "Count") +
+   theme_minimal() +
+   scale_fill_manual(values = c("Positive" = "green", "Neutral" = "gray", "Negative" = "red"))
> # Bar chart: Perceived Impact of AI Tools
> ggplot(data, aes(x = Impact_of_AI_Tools, fill = Impact_of_AI_Tools)) +
+   geom_bar() +
+   labs(title = "Perceived Impact of AI Tools on Academic Performance",
+        x = "Perceived Impact",
+        y = "Count") +
+   theme_minimal() +
+   scale_fill_manual(values = c("Positive" = "green", "Neutral" = "gray", "Negative" = "red"))
>

```

Data	
data	100 obs. of 6 variables
\$ Time_on_Social_Media_Min_per_day	: int 132 136 101 218 50 132 151 244 117 129 ...
\$ Time_Spent_Using_AI_Tools_Hrs_per_week	: num 0 6 3 5 9 2 4 8 2 1 ...
\$ Study_Hours_per_day	: num 6 2 2 7 5 1 2 6 1 2 ...
\$ GPA	: num 2.1 3.06 3.08 3.27 3.45 3.95 3.03 2.65 3.59 2.54 ...
\$ Impact_of_Social_Media	: chr "Positive" "Negative" "Neutral" "Negative" ...
\$ Impact_of_AI_Tools	: chr "Neutral" "Neutral" "Positive" "Negative" ...
mean_gpa_ai	11 obs. of 2 variables
\$ Time_Spent_Using_AI_Tools_Hrs_per_week	: num 0 1 2 3 4 5 6 7 8 9 ...
\$ GPA	: num 2.56 3.03 3.34 2.9 2.98 ...
mean_gpa_social	85 obs. of 2 variables
\$ Time_on_Social_Media_Min_per_day	: int 31 34 38 43 44 50 51 57 62 64 ...
\$ GPA	: num 2.87 3.55 3.81 2.23 2.81 ...
summary_stats	4 obs. of 4 variables
\$ Variable:	chr "Time on Social Media (min/day)" "Time Using AI Tools (hrs/week)" "Study Hours (hrs/day)..."
\$ Mean	: num 168.14 4.9 4.69 2.98
\$ Median	: num 166.5 5 5 3.02
\$ SD	: num 81.899 3 2.102 0.587
Values	
cor_ai_gpa	-0.0176038704379821
cor_social_gpa	-0.096531191981381
cor_study_gpa	-0.213641074319596

Here is the link for the R file:

https://drive.google.com/file/d/1Iqg7tyIFds_YXrs09HJlMMx_olTVpG_V/view?usp=sharing

Appendix C: Data table

	Time_on_Social_Media_Min_per_day	Time_Spent_Using_AI_Tools_Hrs_per_week	Study_Hours_per_day	GPA	Impact_of_Social_Media	Impact_of_AI_Tools
1	132	0	6	2.10	Positive	Neutral
2	136	6	2	3.06	Negative	Neutral
3	101	3	2	3.08	Neutral	Positive
4	218	5	7	3.27	Negative	Negative
5	50	9	5	3.45	Neutral	Neutral
6	132	2	1	3.95	Negative	Positive
7	151	4	2	3.03	Neutral	Negative
8	244	8	6	2.65	Positive	Negative
9	117	2	1	3.59	Negative	Negative
10	129	1	2	2.54	Positive	Neutral
11	181	3	5	2.88	Neutral	Neutral
12	160	2	6	2.16	Negative	Positive
13	179	9	6	2.05	Negative	Negative
14	287	8	3	3.93	Neutral	Negative
15	221	6	6	3.67	Neutral	Positive
16	190	9	3	3.39	Neutral	Positive
17	51	8	3	2.82	Positive	Positive
18	282	2	6	2.35	Neutral	Negative
19	265	9	6	2.31	Neutral	Negative
20	78	5	7	2.50	Negative	Positive
21	88	8	6	3.10	Negative	Neutral

Showing 1 to 22 of 100 entries, 6 total columns

Appendix D: Time spent using AI Tools vs GPA

	Time_Spent_Using_AI_Tools_Hrs_per_week	GPA
1	0	2.558333
2	1	3.032500
3	2	3.337857
4	3	2.901667
5	4	2.980000
6	5	2.753000
7	6	2.932222
8	7	3.118571
9	8	3.068889
10	9	3.034667
11	10	2.440000

Appendix E: Time on Social Media vs GPA

	Time_on_Social_Media_Min_per_day	GPA
1	31	2.870
2	34	3.550
3	38	3.810
4	43	2.230
5	44	2.815
6	50	3.080
7	51	2.820
8	57	2.050
9	62	3.390
10	64	2.850
11	70	3.390
12	71	3.500
13	73	3.250
14	77	3.260
15	78	2.500
16	79	2.350
17	80	3.480
18	82	3.430
19	83	2.800
20	84	3.110

Showing 1 to 20 of 85 entries, 2 total columns

Appendix F: Data Summaries Table

← → 📄 Filter				
	Variable	Mean	Median	SD
1	Time on Social Media (min/day)	168.1400	166.50	81.8985601
2	Time Using AI Tools (hrs/week)	4.9000	5.00	3.0000000
3	Study Hours (hrs/day)	4.6900	5.00	2.1019231
4	GPA	2.9823	3.02	0.5866096